

Part 5: The Schwinger Mechanism and Color Flux Tubes

How Pair Production in a Strong Field Motivates the String Picture

PHYS 7502 – Final Reading Assignment

1 The Question Behind This Chapter

Here's a concrete question that will drive this whole part: *when you pull two quarks apart, what happens physically to the color field between them?*

In QED, if you pull two charges apart, the electric field lines spread out in all directions – Coulomb's law, straight out of E&M 1. The potential energy falls as $1/r$. This is what you'd naively guess for QCD too.

But QCD is *not* QED. The gluons carry color charge, and they interact with each other (the non-Abelian property). This changes everything about what the color field does at long distances. Instead of spreading out, it gets *compressed* into a narrow tube.

Why a Tube, Not a Spread-Out Field?

A quick way to see why gluon self-interactions give flux tubes: the gluon attracts itself. Field lines that would otherwise spread apart get pulled back together by the gluon self-interaction. The energy minimum isn't a Coulomb-like spread-out configuration, but a localized tube-shaped one. Lattice QCD simulations have confirmed this picture in detail.

This chapter uses a simpler, *analytic* model – the Schwinger mechanism – to understand what happens in such a tube when the energy stored in it becomes large enough to create a new quark-antiquark pair. It's simpler than full QCD, but the physical intuition it builds transfers directly.

2 The Schwinger Mechanism in QED: Pair Production in a Field

Julian Schwinger, in 1951, asked a beautiful question: if you put a very strong, uniform electric field into vacuum, will electron-positron pairs be produced?

The answer is **yes**, and he computed the rate. The mechanism:

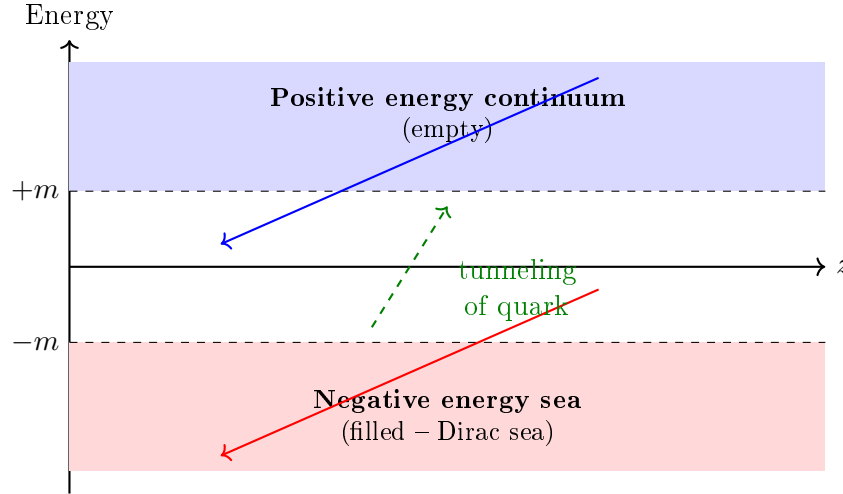


Figure 1: The Schwinger picture of pair production, drawn in Dirac’s hole-theory language. The uniform electric field tilts both the positive-energy continuum and the negative-energy sea. A particle in the negative-energy sea at some z can *tunnel* across the forbidden gap (of size $2m$) and emerge in the positive-energy continuum at a different z . The hole it leaves behind is an antiparticle. Net result: a particle-antiparticle pair is created.

Schwinger Pair Production, Physical Picture

In a strong uniform electric field $\mathcal{E}$, the vacuum is unstable: pairs are continuously being created by *quantum tunneling*. An “electron” sitting in the negative-energy sea has total energy $-mc^2$; an external field can tip the energy scales so that, at some point in space, the negative-energy level matches up with a positive-energy level on the other side. The particle can quantum-mechanically tunnel across the gap. The hole it leaves behind is a positron, which drifts off in the opposite direction.

2.1 The famous result

Using the WKB approximation for the tunneling probability, Schwinger found:

Schwinger’s Pair Production Rate

In a strong electric field $\mathcal{E}$ uniform over a volume V , the rate of pair production per unit volume per unit time is

$$\frac{dN}{dt d^3x} = \frac{(e\mathcal{E})^2}{8\pi^3} \exp\left[-\frac{\pi m^2 c^3}{e\mathcal{E}\hbar}\right] \quad (1)$$

The key feature is the **exponential suppression** by the electron mass: if the field isn’t strong enough that $e\mathcal{E}/m \gtrsim mc^3/\hbar$, pair production is negligible.

In ordinary lab electric fields, this exponent is astronomically large, so pair production is never seen – you need *enormous* fields. But the physics is real, and it transfers directly to QCD.

3 Translating to QCD: the Color Flux Tube

Now comes the step that this chapter is really about. We *replace* the electromagnetic field in Schwinger’s problem with a model of the color field between a quark and an antiquark.

3.1 The setup

Put a quark q_0 at $z = 0$ and an antiquark $\bar{q}_0$ at $z = L$, as in Figure 2. Assume the color field between them is confined to a **flux tube** – a narrow cylinder of cross-sectional area A – with uniform color electric field $\vec{\mathcal{E}}$ pointing along $\hat{z}$. Outside the tube, the field is zero.

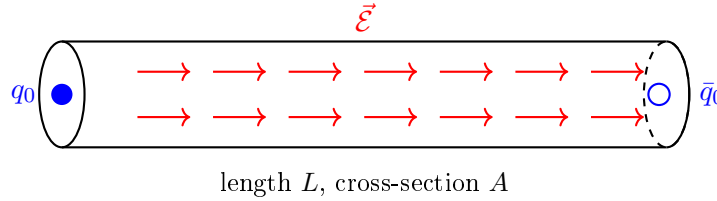


Figure 2: The color flux tube model. A quark q_0 and an antiquark $\bar{q}_0$ are separated by a distance L . The color electric field $\vec{\mathcal{E}}$ between them is confined to a narrow tube of cross-sectional area A (typically a circle of radius ~ 0.5 fm). Outside the tube, the field is zero.

3.2 The energy content

The field energy density is $\frac{1}{2}\mathcal{E}^2$, so the total energy in the tube is:

$$U_{\text{field}} = \frac{1}{2}\mathcal{E}^2 \cdot A \cdot L \quad (2)$$

This is the interaction energy between q_0 and $\bar{q}_0$.

The String Tension

The interaction energy is *linearly proportional* to the separation L :

$$U(L) = \kappa \cdot L \quad \text{where} \quad \kappa = \frac{1}{2}\mathcal{E}^2 A \quad (3)$$

The constant κ is called the **string tension**. You can estimate κ from the Regge slope (Part 7) and find

$$\kappa \approx 1 \text{ GeV/fm} \quad (4)$$

This linear potential is the key feature: as you try to pull the quarks apart, the energy stored in the field grows without bound. Compare this to QED, where the Coulomb energy levels off at a finite value ($-ke^2/r \rightarrow 0$ as $r \rightarrow \infty$). The linearly-rising potential is the origin of confinement.

3.3 Relating $\mathcal{E}$, κ , and the color charge

By Gauss’s law applied to one end of the tube:

$$\mathcal{E} \cdot A = q, \quad (5)$$

where q is the color charge of the quark. Multiplying both sides of equation (3) by 2 and using Gauss's law:

$$q\mathcal{E} = 2\kappa \quad (6)$$

(This relation will be useful soon when we compute the force on a newly produced quark.)

4 Pair Production in the Flux Tube: the Qualitative Picture

Now here's where Schwinger's mechanism enters. Imagine the tube is stretched to some length L , and the field $\mathcal{E}$ inside it is strong enough to cause pair production. A new $q_1\bar{q}_1$ pair pops into existence somewhere in the middle:

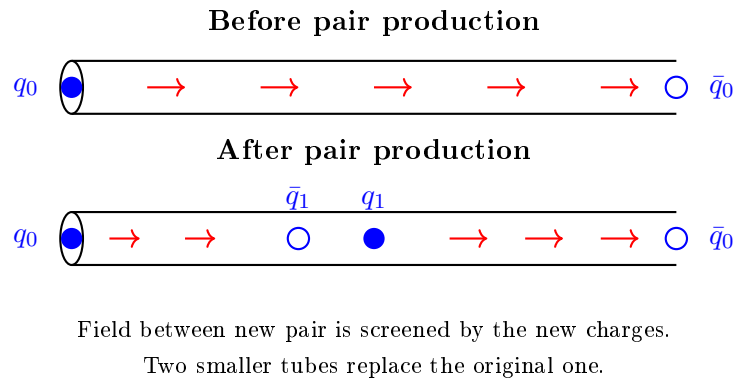


Figure 3: Pair production in the flux tube. When a $q_1\bar{q}_1$ pair is created, the new antiquark $\bar{q}_1$ pairs up with the original quark q_0 on the left, and the new quark q_1 pairs up with the original antiquark $\bar{q}_0$ on the right. What was one long string is now two shorter ones. This is the microscopic picture of hadron production.

Why Pair Production Matters for Hadronization

This picture is **exactly** what hadronization looks like in the string model. When you try to pull two quarks apart:

1. The color field between them forms a tube (flux tube).
2. As you pull, the tube stretches and the energy in it grows linearly.
3. At some point, the energy in the tube is enough to create a new $q\bar{q}$ pair.
4. The new pair “breaks” the tube: $q\bar{q}_{\text{new}}$ on one side, $q_{\text{new}}\bar{q}$ on the other, each now bound by a shorter tube of lower energy.
5. Each of those shorter tubes is a **meson**.

So the mechanism by which a lone quark turns into hadrons is: **string breaking by Schwinger-type pair production**.

5 The Quantum Mechanics of the Tunneling

Let me sketch the actual calculation of the pair production rate in the flux tube. (The instructor said to skim most of this – the essential physics is the picture above, and the result below.)

5.1 Setting up the potential

Treat the $q_0\bar{q}_0$ pair as fixed external color charges generating a constant $\mathcal{E}$ inside the tube. A newly produced quark q_1 with charge q inside the tube would experience a potential

$$V(z) = -q\mathcal{E}z = -2\kappa z \quad (7)$$

(using $q\mathcal{E} = 2\kappa$ from eq. 6). However, the *newly produced antiquark* $\bar{q}_1$ is always produced nearby, and its field partially cancels the original. The net potential seen by q_1 works out to

$$V(z) = -\kappa z \quad (8)$$

That is, q_1 feels the string tension κ , as it should.

5.2 The tunneling barrier

Using the Dirac (or Klein-Gordon) equation for a relativistic quark of mass m in this potential, and reducing to an effective Schrödinger-like equation by standard tricks (the WKB method), you find an effective barrier for pair production between the forbidden and allowed regions:

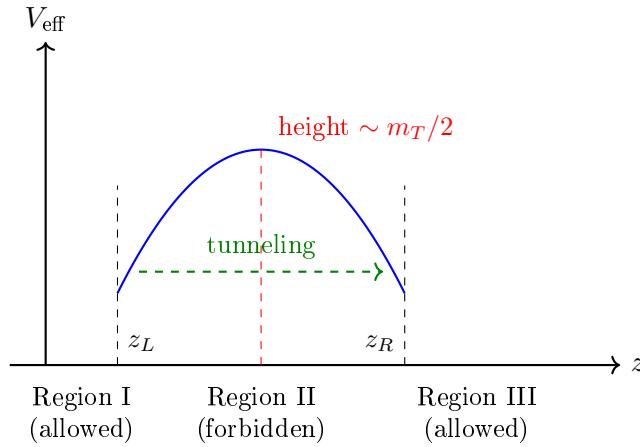


Figure 4: Effective barrier for quark-antiquark pair production in the flux tube. The quark state lies in Region I, the tunneling barrier (of height $\sim m_T/2$ where $m_T = \sqrt{m^2 + p_T^2}$) extends from z_L to z_R , and emergence into Region III produces a particle-antiparticle pair.

5.3 The rate, à la WKB

Standard WKB gives a tunneling probability

$$P = \exp \left[- \int_{z_L}^{z_R} 2\sqrt{2m_{\text{eff}} [V_{\text{eff}}(z) - E_{\text{eff}}]} dz \right] \quad (9)$$

Working through the details, this yields the headline result:

Tunneling Probability for $q\bar{q}$ Pair Production

$$P = \exp \left[-\frac{\pi(m^2 + p_T^2)}{\kappa} \right] \quad (10)$$

where m is the quark's mass and p_T its transverse momentum. This is the QCD analog of Schwinger's original formula.

6 Physical Consequences of Equation (10)

Equation (10) is a compact, information-rich formula. Let's read off what it tells us.

6.1 Heavier quarks are harder to produce

Since $P \sim e^{-\pi m^2/\kappa}$:

- For up/down quarks: $m \approx 5 \text{ MeV}$, so $\pi m^2/\kappa \approx 10^{-4}$. Production is *not* suppressed meaningfully.
- For strange quarks: $m \approx 150 \text{ MeV}$, so $\pi m^2/\kappa \approx 0.1$. A factor $e^{-0.1} \approx 0.9$ suppression – small but measurable.
- For charm quarks: $m \approx 1.3 \text{ GeV}$, so $\pi m^2/\kappa \sim 5$. $e^{-5} \approx 0.007$: effectively forbidden.
- For bottom: totally forbidden by this mechanism.

Heavy-Quark Suppression

The Schwinger mechanism *predicts* heavy-quark suppression in hadronization. This is observed! Hadron yields in jets are dominated by pions, followed by kaons (with a suppression of $\sim 30\%$ per strange quark), followed by far fewer charm/bottom hadrons. The pattern is consistent with exponential suppression in mass.

This is why PYTHIA (which uses the Lund string model, a generalization of what we're discussing) has a parameter for the “strangeness-suppression factor” γ_s that gets tuned to data – the formula gives you roughly the right magnitude, but the detailed number needs data to pin down.

6.2 Transverse momentum distribution

The $e^{-\pi p_T^2/\kappa}$ factor in equation (10) is a Gaussian in p_T . So the mechanism predicts that produced particles have transverse momenta distributed as

$$\frac{dN}{dp_T^2} \propto e^{-\pi p_T^2/\kappa} \quad (11)$$

with a width

$$\langle p_T^2 \rangle \sim \frac{\kappa}{\pi} \sim 0.3 \text{ GeV}^2 \quad (12)$$

i.e., $\langle p_T \rangle \sim 500 \text{ MeV}$. This matches the observed “primordial transverse momentum” of hadrons in jets remarkably well.

6.3 Minimum string length for pair production

For pair production to be kinematically allowed, the initial string has to be long enough to provide the rest mass of the produced pair. Setting the threshold condition gives:

$$L_{\min} = \frac{2m_T}{\kappa} \quad (13)$$

For light quarks at rest: $L_{\min} \approx 2 \cdot 0.33 \text{ GeV}/(1 \text{ GeV}/\text{fm}) \approx 0.7 \text{ fm}$. For a quark with transverse momentum $p_T = 0.35 \text{ GeV}$, it grows to about 1 fm. These are reasonable hadronic scales.

7 The Bigger Picture: From Schwinger to the String Model

Everything you've just read is the conceptual foundation for the Lund string model (Part 6). Here's the logical flow:

1. **Non-Abelian QCD $\rightarrow$ flux tubes.** Gluon self-interaction compresses the color field into tubes. Lattice QCD confirms this.
2. **Linear potential.** Energy in a tube grows linearly with length: $U(L) = \kappa L$.
3. **Schwinger mechanism.** If a tube is stretched enough, the color electric field produces new $q\bar{q}$ pairs by quantum tunneling.
4. **String breaking.** The new pair breaks the tube into two shorter tubes, producing mesons.
5. **Hadronization = iterated string breaking.** As a parton moves away from a hard process, it drags a string behind it. The string stretches, breaks, stretches, breaks – each break produces a hadron.

What You Should Take Forward

The specific formulas in this chapter (WKB derivation, etc.) are less important than the *physical picture*:

- Confinement $\Leftrightarrow$ flux tubes with linear potential κr .
- Hadronization $\Leftrightarrow$ strings breaking via Schwinger-type pair production.
- Heavy-quark suppression and primordial p_T fall out naturally from the exponential form $e^{-\pi(m^2 + p_T^2)/\kappa}$.
- The string tension $\kappa \approx 1 \text{ GeV}/\text{fm}$ is the **one** parameter that ties everything together.

Everything in Part 6 (the classical string model) and Part 7 (DPM, Regge, Pomeron) is built on these ideas.

8 Summary of Part 5

Main Takeaways

1. In QCD, gluon self-interactions squeeze the color field between a $q\bar{q}$ pair into a **flux tube** of roughly constant cross-section.
2. The energy stored in a flux tube of length L is $U(L) = \kappa L$, with string tension $\kappa \approx 1 \text{ GeV/fm}$. This **linear potential** is the origin of confinement.
3. The **Schwinger mechanism** (originally for QED pair production) carries over to QCD: strong enough color fields produce $q\bar{q}$ pairs by quantum tunneling.
4. The production probability $P \propto \exp[-\pi(m^2 + p_T^2)/\kappa]$ has two immediate consequences: **heavy-quark suppression** and a **Gaussian p_T distribution** for produced hadrons – both observed.
5. Hadronization proceeds by repeated **string breaking**: a stretched color flux tube snaps, producing new $q\bar{q}$ pairs, which form mesons.
6. This entire picture is the conceptual foundation of the Lund string model (next part) and of how PYTHIA implements hadronization.